

METODA MATRICILOR DE TRANSFER. IMPLEMENTAREA CALCULULUI S5

$$ee := 1.6 \cdot 10^{-19} \quad hplank := 6.6260755 \cdot 10^{-34} \quad \underline{m} := 9.1093897 \cdot 10^{-31}$$

$$h := \frac{hplank}{2 \cdot \pi}$$

n := 4 Numarul de puncete de discontinuitate a potentialului:este 5. Indexarea elementelor incepe de zero

i := 0 .. n Definirea barierei de potential:

$$Vj_i :=$$

0
0.2
0
0.2
0

lnm_i := Largimea barierei de potential

5
5
5
5
5

Rezultate la energia E0, la care am efectuat cu mana calculul la seminar

$$\underline{l}_i := lnm_i \cdot 10^{-9}$$

$$E0 := 0.0602471$$

Vectorul de unda:

$$t := \frac{2 \cdot m \cdot ee}{h^2}$$

$$t = 2.62111597 \times 10^{19} \quad m^{-2} \cdot eV$$

$$k(i, E) := \sqrt{t \cdot (E - Vj_i)}$$

$$k(0, E0) = 1.2566409 \times 10^9$$

$$k(1, E0) = 1.91391891i \times 10^9$$

$$\frac{k(2, E0)}{k(1, E0)} = -0.65658001i$$

$$r(i, E) := \frac{E - Vj_i}{E - Vj_{i-1}}$$

$$r(1, E0) = 1.52304362i$$

$$r(3, E0) = 1.52304362i$$

$$r(2, E0) = 0.65658001i$$

$$r(4, E0) = 0.65658001i$$

Definim matricile de transfer si transport

$$M_{transfer}(i, E) := \frac{1}{2} \cdot \begin{pmatrix} 1 + r(i, E) & 1 - r(i, E) \\ 1 - r(i, E) & 1 + r(i, E) \end{pmatrix} \quad M_{transfer}(1, E_0) = \begin{pmatrix} 0.5 + 0.76152181i & 0.5 - 0.76152181i \\ 0.5 - 0.76152181i & 0.5 + 0.76152181i \end{pmatrix}$$

$$M_{transfer}(2, E_0) = \begin{pmatrix} 0.5 + 0.32829001i & 0.5 - 0.32829001i \\ 0.5 - 0.32829001i & 0.5 + 0.32829001i \end{pmatrix}$$

$$M_{transport}(i, E) := \begin{pmatrix} e^{-ik(i, E) \cdot l_i} & 0 \\ 0 & e^{ik(i, E) \cdot l_i} \end{pmatrix}$$

$$M_{transport}(1, E_0) = \begin{pmatrix} 1.43226078 \times 10^4 & 0 \\ 0 & 6.98196875 \times 10^{-5} \end{pmatrix}$$

$$M_{transport}(2, E_0) = \begin{pmatrix} 1 - 1.9208346i \times 10^{-5} & 0 \\ 0 & 1 + 1.9208346i \times 10^{-5} \end{pmatrix}$$

$$MP(i, E) := M_{transfer}(i, E) \cdot M_{transport}(i, E)$$

$$PROD(E) := \prod_{i=1}^{n-1} MP(i, E)$$

$$M(E) := PROD(E) \cdot M_{transfer}(n, E)$$

$$M(E_0) = \begin{pmatrix} 1.00000001 + 2.33995334i \times 10^3 & 2.14711688 \times 10^3 + 930.19930376i \\ 2.14711688 \times 10^3 - 930.19930376i & 1.00000001 - 2.33995334i \times 10^3 \end{pmatrix}$$

$$T(E) := \left(\left| M(E)_{0,0} \right| \right)^{-2}$$

$$T(E_0) = 1.82635638 \times 10^{-7}$$

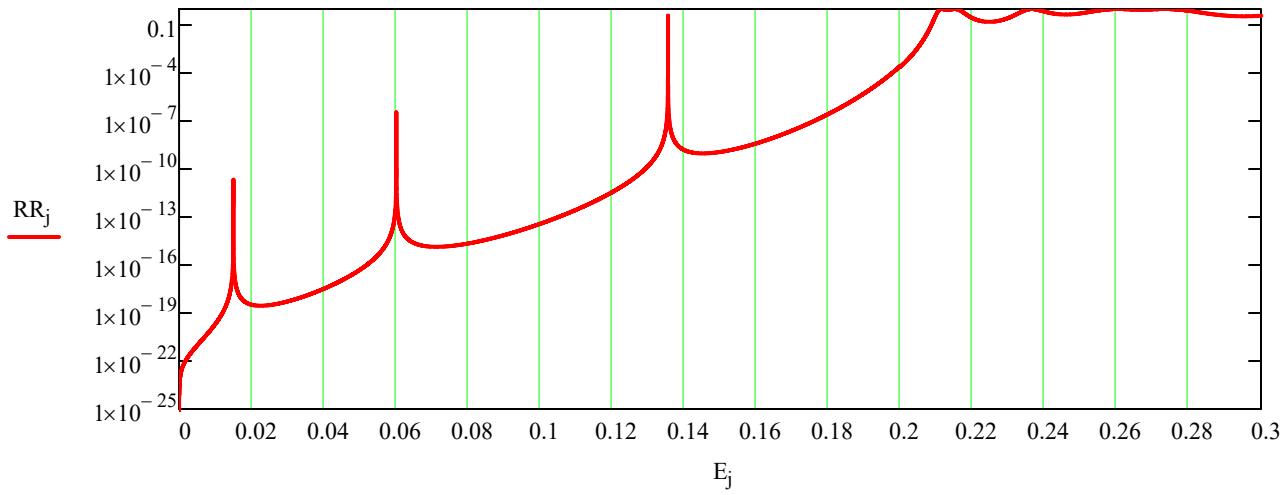
Reprezentare grafica a transmitantei

$$j := 1 \dots 300000$$

$$E_j := 0.000001 \cdot j + 0.000001$$

$$\max(E) = 0.300001$$

$$RR_j := T(E_j)$$



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rezultat := submatrix(augment(E,RR),1,rows(RR) - 1,0,1)
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rezultat =

	0	1
0	$2 \cdot 10^{-6}$	0
1	$3 \cdot 10^{-6}$	0
2	$4 \cdot 10^{-6}$	0
3	$5 \cdot 10^{-6}$	0
4	$6 \cdot 10^{-6}$	0
5	$7 \cdot 10^{-6}$	0
6	$8 \cdot 10^{-6}$	0
7	$9 \cdot 10^{-6}$	0
8	$1 \cdot 10^{-5}$	0
9	$1.1 \cdot 10^{-5}$	0
10	$1.2 \cdot 10^{-5}$	0
11	$1.3 \cdot 10^{-5}$	0
12	$1.4 \cdot 10^{-5}$	0
13	$1.5 \cdot 10^{-5}$	0
14	$1.6 \cdot 10^{-5}$	0
15	$1.7 \cdot 10^{-5}$	...

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stari :=
| s ← -1
| for i ∈ 1 .. 200000 - 2
|   if (rezultat⟨1⟩)i-1 < (rezultat⟨1⟩)i > (rezultat⟨1⟩)i+1
|     | s ← s + 1
|     | auxs ← (rezultat⟨0⟩)i
| aux
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$$\text{stari} = \begin{pmatrix} 0.015062 \\ 0.060247 \\ 0.135555 \\ 0.2 \end{pmatrix}$$

$$\frac{h^2 \cdot \pi^2}{2 \cdot m \cdot 5^2 \cdot (10^{-9})^2 \cdot 1.6 \cdot 10^{-19}} = 0.01506168$$

$$\frac{h^2 \cdot \pi^2 \cdot 4}{2 \cdot m \cdot 5^2 \cdot (10^{-9})^2 \cdot 1.6 \cdot 10^{-19}} = 0.06024673$$

$$\frac{h^2 \cdot \pi^2 \cdot 9}{2 \cdot m \cdot 5^2 \cdot (10^{-9})^2 \cdot 1.6 \cdot 10^{-19}} = 0.13555515$$